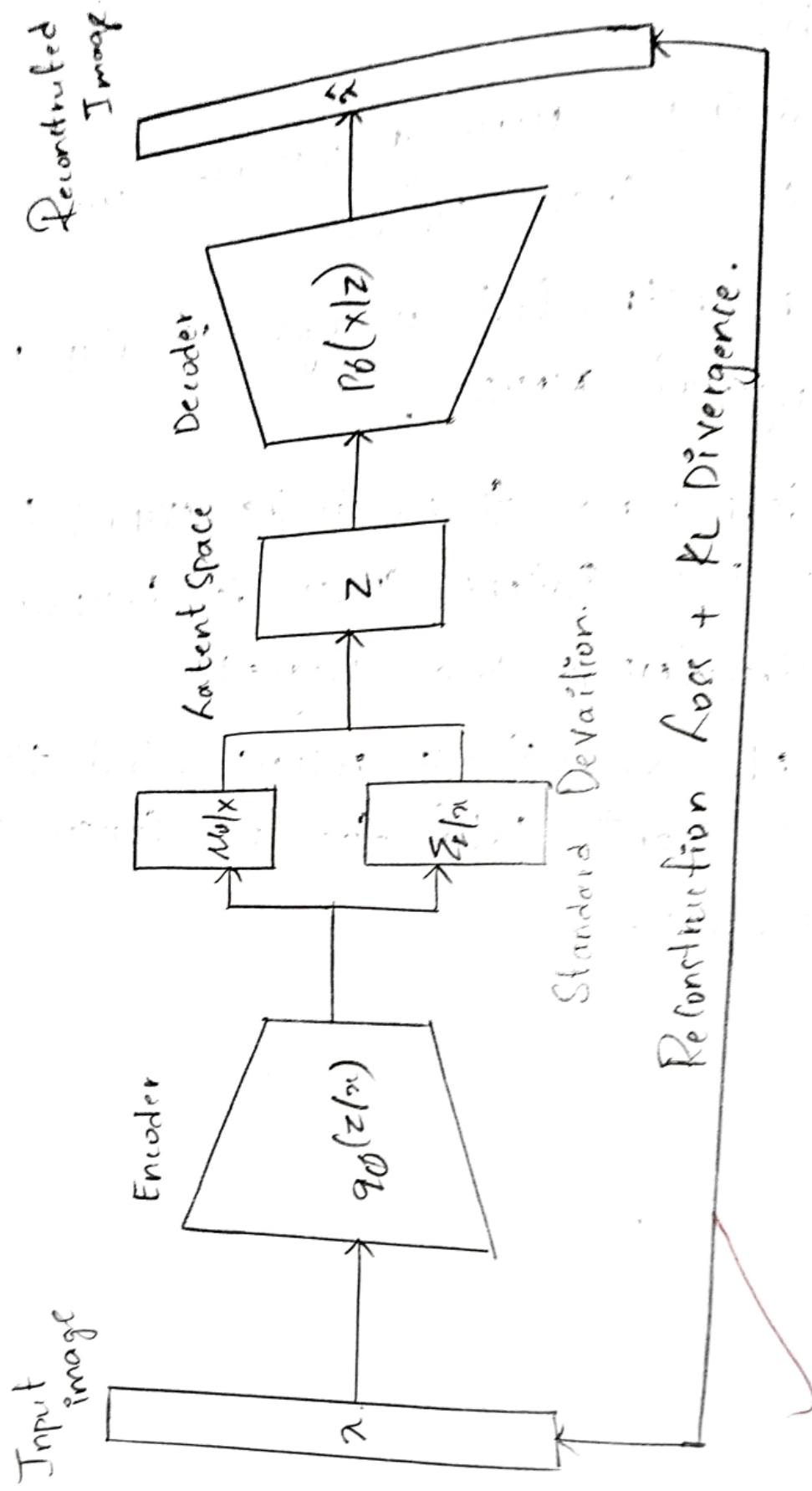


Variational Autoencoder Architecture.



Lab: 11 Experiments Using Variational Autoencoder

Aim: To implement and analyze a Variational Autoencoder (VAE) model on MNIST dataset for generating new digit images by learning a Probabilistic latent space representation.

Algorithm:-

- Load & Preprocess the MNIST dataset
 - * Normalize pixel values between 0 and 1
 - * Reshape input (28x28x1) for CNN-based model
- Build the Encoder Network
 - * Input 28x28x1 image
 - * Apply conv layer to extract features
 - * Flatten and use dense layer to produce two outputs.
- Perform Reparameterization Trick.
 - * Sample latent vector $z = \mu + \sigma \cdot \epsilon$, $\epsilon \sim N(0,1)$
- Build a Decoder Network.
 - * Input: latent vector z
 - * Use Dense & Conv layers
- Define VAE Loss function
- Compile the model using an optimizer.
- Train the VAE
- Evaluate the model.

Observation:

Epoch 1: Avg loss 110.8988

Avg test loss 109.8923

Epoch 2: Avg loss 110.4137

Avg test loss 109.7505

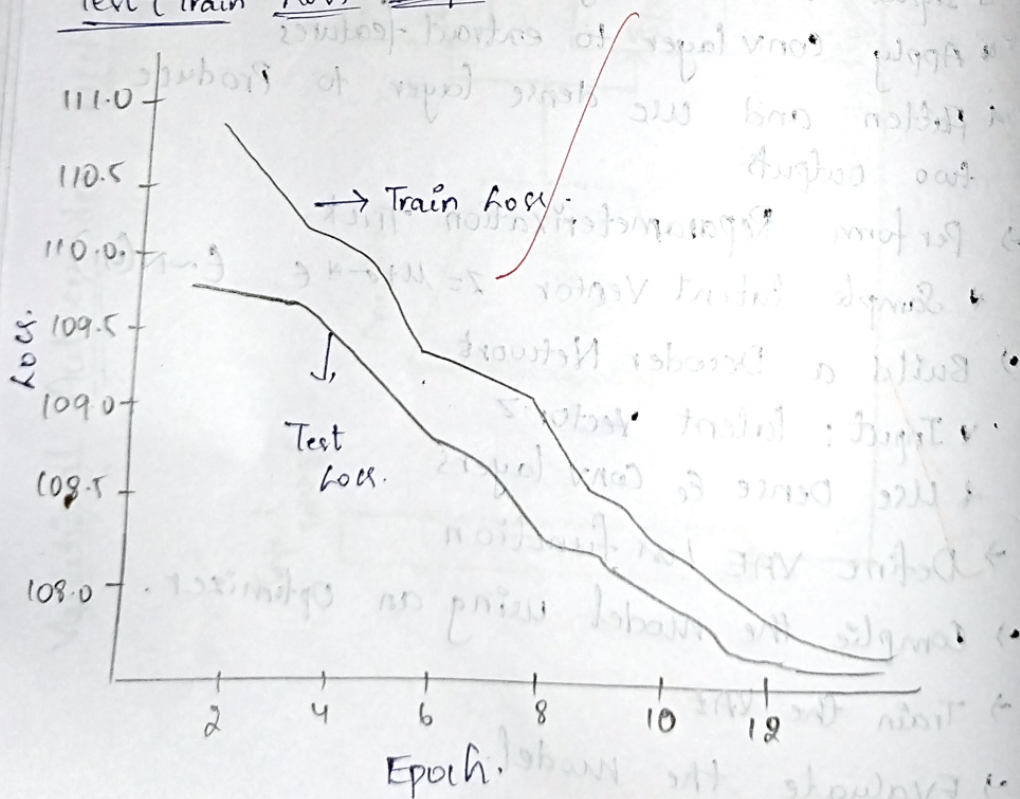
Epoch 3: Avg loss 109.3489

Avg test loss 108.5792

Epoch 10: Avg train loss 108.0769

Avg test loss 107.6855

Test (Train) Loss Graph



→ After training

- Set model to
- Take a batch
- get reconstructions
- plot original
- side by side.

Observation:

→ Instead of mapping
VAE encode input
in latent space

→ VAE's use the
for backpropagation
process.

→ The loss function
loss & KL divergence
to follow a prior

Result:

Successful

~~Autoencoder~~

→ After training

- Set model to eval model
- Take a batch from test data
- get reconstructed images
- plot original & reconstructed images
- side by side.

Observation:

- insted of mapping input to a fixed point, VAE encode input into a Probability distribution in latent space
- VAE's use the reparameterization trick to allow for backpropagation through the sampling process.
- The loss function of VAE includes a Reconstruction loss & KL Divergen to regularize space & encourage to follow a prior distribution.

Result:

Successfully implement VAE (Variational Autoencoders) using MNIST dataset.